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Journal of Manufacturing Processes

Date: 15 September 2025

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Creating elemental gradient transition layer to improve the thermal fatigue resistance of femtosecond laser glass-metal welding

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Abstract

High quality glass-metal weld seam necessitate robust welding strength and superior in-service performance. However, the mismatch of thermal properties in heterogeneous materials leads to cracks at the bonding interface due to environmental temperature fluctuations. This study focuses on the thermal fatigue resistance requirements of glass-metal femtosecond laser welding, investigates the elemental migration mechanism at the interface of heterogeneous materials. A novel welding strategy based on plasma channels creation to form a gradient transition layer of elemental concentration at the interface is proposed. To enhance the diffusion height of metallic elements, it is suggested to refine the welding process by incorporating a tin-based solder layer with low ablation threshold between the glass and metal components. With the implementation of the optimized

Outline

[Abstract](#)[Keywords](#)[1. Introduction](#)[2. Experimental details](#)[3. Results and discussion](#)[4. Conclusion](#)[CRediT authorship contribution statement](#)[Declaration of competing interest](#)

strategy, welding quartz glass to copper (CTE, quartz: copper=1:33.4) achieves an initial welding strength of 34.9 Mpa. After undergoing 100 thermal cycles ranging from -40°C to 150°C , in comparison to samples without a gradient transition layer, the attenuation of welding strength significantly decreased from 95.1% to 29.8%. This study offers a fresh perspective on improving the thermal fatigue resistance of glass-metal femtosecond laser welding, revealing significant potential for advancing the development of high-performance glass-metal welding technology.

Keywords

Glass-metal welding; Femtosecond laser; Thermal fatigue resistance; Thermal cycles

[Previous article in this issue](#)

[Next article in this issue](#)

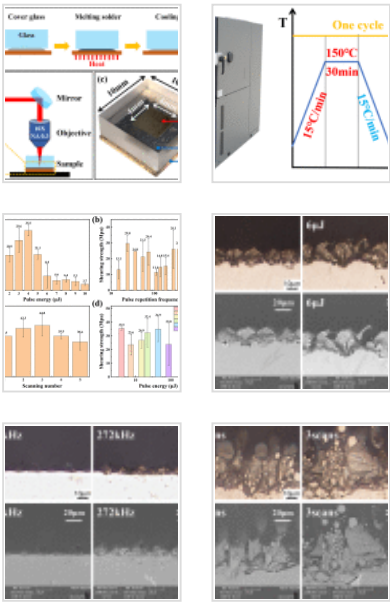
1. Introduction

Glass's excellent chemical and physical properties make it an excellent material for high-end optical systems, astronomy and satellites, microelectronics and life sciences [1,2]. However, glass has poor mechanical properties due to its brittleness and low impact resistance, which severely limits its use in engineering applications. By connecting glass with metals, which have good elasticity and high impact resistance, the overall mechanical performance can be significantly improved. As a result, the application of glass-metal bonding in micro-electro-mechanical systems, medical implants, optical sensors and other fields has received widespread attention [3,4].

High quality glass-metal weld seam necessitate robust welding strength, minimal welding defects, and superior in-service performance. However, due to the significant disparities in both physical and chemical properties between glass and metal, traditional joining methods such as active brazing [5], laser welding [6], matching sealing [7], anodic

- Acknowledgements
- Data availability
- References
- Show full outline

Figures (17)



[Show 11 more figures](#)

Tables (3)

- Table 1. Melting points and ablation thre...
- Table 2. The laser parameters of three w...
- Table 3. Comparison of elemental transit...

bonding [8], adhesive bonding [9], among others, struggle to achieve a highly reliable bonding between the two materials. Vacuum brazing of glass and metal predominantly relies on metal solders that contain active elements. The types of metal used are typically limited to a narrow range, such as stainless steel and Kovar alloy. The lengthy welding times and high temperatures involved significantly impede its practical application [10]. In conventional laser fusion welding, glass and metal can be joined by applying localized heat to the metal. However, this method is unsuitable for glass bonding due to the extensive heat-affected zone (HAZ) that results from the large thermal input [11]. Matching sealing often involves pre-oxidizing the metal to form an oxide layer. The primary drawback of matching sealing is its lengthy production cycle and the relatively weak bonding strength [12,13]. Anodic bonding is a technique that utilizes the combined effects of heat and an electric field. The main limitation of anodic bonding is the restricted range of materials that can be bonded, as the material combinations must have similar coefficients of thermal expansion (CTE) [14]. Adhesive bonding is a straightforward and practical approach, though it suffers from low precision, poor environmental adaptability, and subpar long-term performance [15,16].

Ultrafast laser induces nonlinear absorption of transparent glass so that both glass and metal can absorb laser energy and realize the fusion connection of heterogeneous materials [17,18]. Numerous studies have explored the strength of ultrafast lasers in bonding glass to metal [19,20]. Through meticulous parameter tuning and beam shaping, researchers have managed to attain consistently robust welding strengths exceeding 20MPa, demonstrating the potential of ultrafast laser for high-strength glass-metal welding [21]. However, the molten pool created by the ultrafast laser is diminutive, which leads to a limited capacity to fill interface gap. Consequently, achieving a superior surface roughness on both the glass and metal is essential to establish an optical contact (with a gap of less than $1/4$ wavelength) [22]. The stringent requirements for welding gaps result in a substantial increase in material costs. More importantly, the reduced size of the molten pool contributes to significant disparities in thermal properties between the two

heterogeneous materials at the interface, which continues to make the issue of poor in-service performance a very serious concern [23].

The mismatch in thermal properties between heterogeneous materials can result in cracks at the welding interface due to temperature fluctuations in the surrounding environment. These cracks can propagate until the connection ultimately fails. The limitations of these weldments, defined by the inherent properties of glass and metal, will significantly reduce their actual service life, thereby limiting their practical industrial applications. Current research indicates that due to the significant difference in CTE between glass and metal (ratio 1:20–60), the thermal cycling process introduces considerable strain at the glass-metal weld seam [24]. Typically, macroscopic cracks form to release some of the stress, which can lead to a significant reduction in welding strength or direct failure [25]. Carter and his team performed only 6 thermal cycling tests, ranging from -50°C to 90°C , on the welded samples created through ultrafast laser welding of metals and glass. Only 5 out of 12 BK7-aluminum alloy welded samples (with a CTE ratio of 1:3.4) did not fail, whereas all 12 quartz glass-aluminum alloy welded samples (with a CTE ratio of 1:47) failed [26]. Hu et al. conducted thermal shock cycling experiments on the femtosecond laser welded samples of sapphire-Si, yet their strength could only be maintained below 5 MPa [27]. In summary, existing research suggests that glass-metal weld seam exhibit poor thermal fatigue resistance, yet there is a lack of specific strategies for improvement. Therefore, it is crucial to develop methods that can significantly improve the thermal fatigue resistance of glass-metal ultrafast laser weld seam while ensuring reliable welding strength.

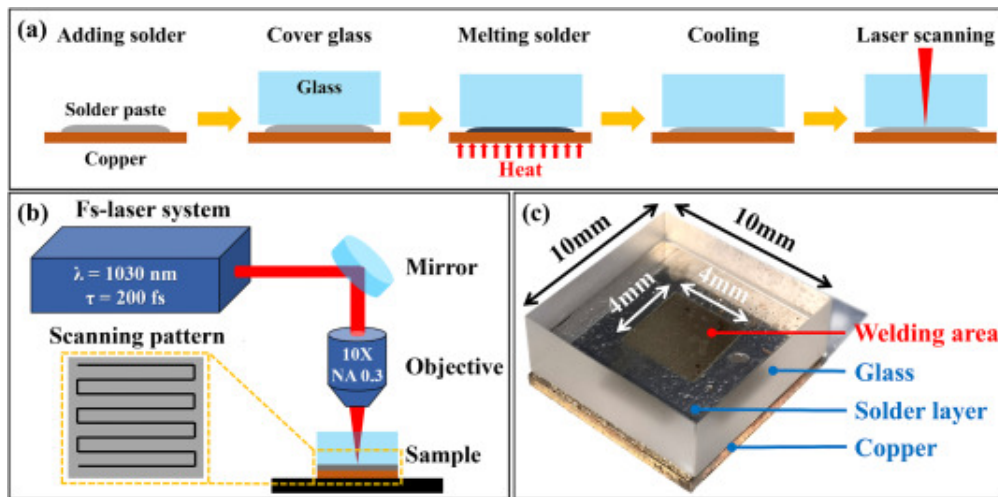
In this study, our commitment is to employ femtosecond laser to tackle the issue of thermal fatigue resistance in glass-metal welding. Considering the significant differences in CTE between glass and metal materials (for instance, quartz: copper=1:33.4, quartz: tin=1:53.4), a novel method is proposed to transport a large amount of metal elements into the interior of glass by constructing a plasma channel within the glass to create a gradient transition layer for matching properties. To enhance the diffusion height of metallic elements, it is suggested to refine the welding process by incorporating a tin-

based solder layer with low ablation threshold between the glass and metal components. This innovative approach aims to enhance the compatibility and bonding strength between the materials. Furthermore, a detailed analysis is conducted to assess the impact of welding parameters on both the welding strength and the weld morphology, and the elemental migration mechanism at the interface of heterogeneous materials under the influence of femtosecond laser is investigated. In conclusion, this research introduces a novel method for improving the thermal fatigue resistance of glass-metal femtosecond laser welding. It ensures minimal strength degradation with increasing temperature cycles, even under high welding strength of 30MPa. The proposed new welding process addresses the shortcomings of existing glass-metal bonding methods, including welding strength, thermal fatigue resistance, and process universality, indicating significant potential for advancing high-performance glass-metal welding technology.

2. Experimental details

As shown in Fig. 1(a), the femtosecond laser welding process employed in this study to join glass to metal primarily encompasses the following five steps: (1) Applying an appropriate thickness of paste solder onto the metal substrate; (2) Positioning the quartz glass atop the paste solder; (3) Utilizing a heating furnace to heat and melt the solder, thereby establishing a brazed joint between the metal and the solder; (4) Allowing the sample to cool down to room temperature; (5) Scanning and welding with a femtosecond laser at the interface of the solder and the glass. This procedure is applicable to the bonding of a variety of metals and glass. In this study, we use copper, a common metal, and quartz glass as examples to conduct our research. The solder is a kind of tin based solder paste (melting point: 227°C, dynamic viscosity: 170±20 pa.s) added with flux (a liquid used to remove oxides on the surface of metal), and its metal composition is 99% tin, 0.3% silver, and 0.7% copper. It is important to highlight that in the third step of process, the solder's very poor wettability for glass means that a brazing connection between the glass and solder will not occur. A solid-state laser (PHAROS PH1-20, Light Conversion) capable of producing laser pulses of $\tau=190\text{fs}$ -10ps duration (200fs is utilized

in this study) at $\lambda = 1030\text{nm}$ wavelength is used in the experiments, as shown in Fig. 1(b). The laser beam is focused into a focal spot (diameter: $15.4\mu\text{m}$) using an objective lens with $10\times$ magnification and a numerical aperture of 0.3. The scanning trajectory is straight lines, with a line spacing of $25\mu\text{m}$. The experimental materials to be welded are quartz glass with the size of $10\text{mm}\times 10\text{mm}\times 4\text{mm}$ and copper with the size of $10\text{mm}\times 10\text{mm}\times 1\text{mm}$, respectively. The solder paste is filled between the glass and the copper with a thickness of $100\mu\text{m}$. The physical appearance of the sample post-welding is depicted in Fig. 1(c), where the weld area is a $4\text{mm}\times 4\text{mm}$ square.



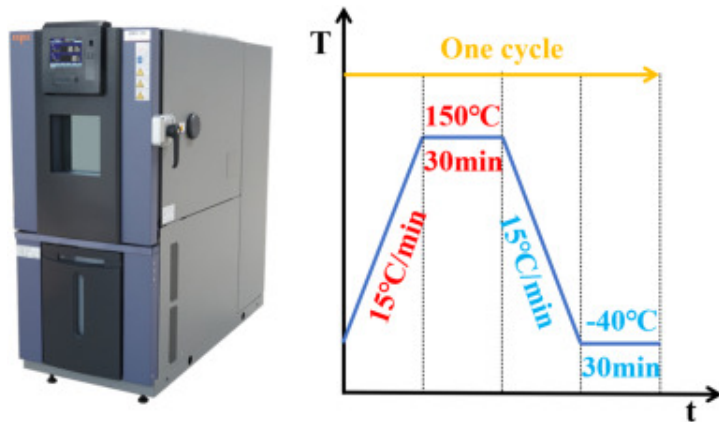
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Fig. 1. (a) Schematic diagram of the femtosecond laser welding process for quartz glass and copper; (b) Schematic diagram of the femtosecond laser welding system and scanning track; (c) Physical image of the quartz glass and copper welded sample.

A tensile tester (HANDPI, HDV-5K) is utilized to evaluate the shearing strength of welded specimens by exerting a tensile force parallel to the weld surface at a controlled tensile speed of 1 mm/min . A minimum of three specimens are subjected to testing for each set of parameters to ascertain the average strength value under each experimental condition.

The shearing strength is calculated by dividing the shearing force by the total welded area. For a detailed analysis of the welding region, the cross-sectional profile of the welded area is analyzed using an optical microscope (OM, OLYMPUS BX 51 m) and a scanning electron microscope (SEM, VEGA3 TESCAN) equipped with energy dispersive spectroscopy (EDS, OXFORD Ultim Extreme). The weld components are further analyzed by X-ray photoelectron spectroscopy (XPS, AXIS Ultra DLD) and X-ray diffractometer (XRD, D8 ADVANCE Da Vinci). Thermal fatigue resistance testing on welded specimens is performed within a rapid temperature change test chamber (ESPEC, GFS-400-15). The physical image of the rapid temperature change test chamber and a flowchart illustrating the temperature cycling process are shown in Fig. 2.



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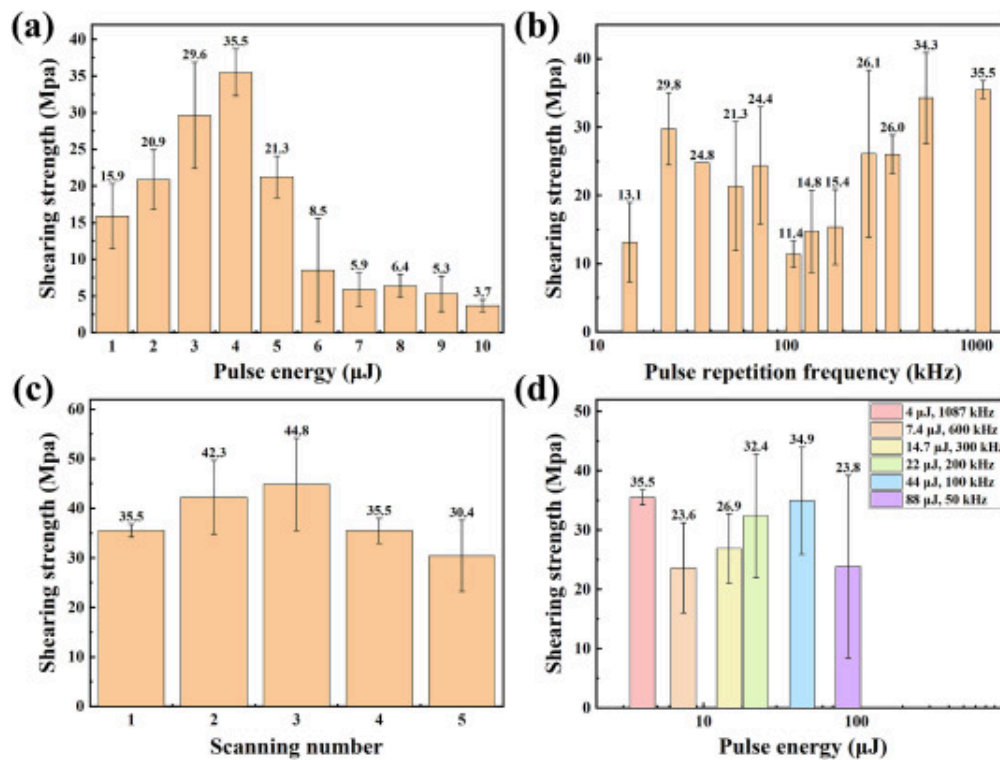
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Fig. 2. The physical image of the rapid temperature change test chamber and a flowchart illustrating the temperature cycling process.

3. Results and discussion

3.1. Influence of femtosecond laser parameters on shearing strength and microstructure of the weld seam

The influence of femtosecond laser pulse energy, pulse repetition frequency and number of repeat scans on the shearing strength and microstructure of the weld seam is analyzed. In addition to the parameters already explained in [Section 2](#), other welding parameters include: scanning speed: 1 mm/s, scanning line spacing 25 μm . Within the energy variable experimental group, the pulse repetition frequency is set at 1087 kHz, with only one single scan being performed. As illustrated in [Fig. 3\(a\)](#), the shearing strength initially rises and subsequently diminishes with increasing pulse energy. Once the pulse energy surpasses 6 μJ , the shearing strength plateaus at a low level with little fluctuation. The peak shearing strength 35.5 Mpa is attained at a pulse energy of 4 μJ .



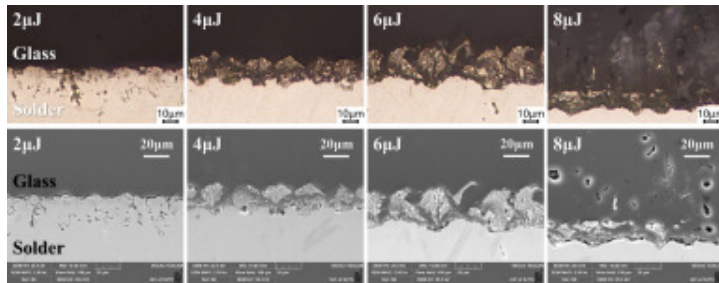
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Fig. 3. Shearing strength of weld joints with varying femtosecond laser parameters. (a) Pulse energy ranging from 1 to 10 μJ ; (b) Pulse repetition frequency ranging from 15 to

1087kHz; (c) Number of repeat scans ranging from 1 to 5; (d) Under the condition of maintaining a constant power of 4.4W, while simultaneously varying the pulse energy and pulse repetition frequency.

The microstructure images of the weld seam captured by OM (upper side) and SEM (lower side) at different pulse energy are shown in [Fig. 4](#). When the pulse energy is 2μJ, the weld morphology exhibits a slightly protruding hill-like appearance. This phenomenon arises because, once the solder and glass melt, the solder's significantly larger CTE compared to that of the glass causes it to expand more in volume. Consequently, the solder extrudes into the glass, creating a convex-concave structure. These numerous convex-concave structures collectively form a serrated weld morphology, which serves as a mechanical anchoring connection. At a pulse energy of 4μJ, the weld morphology exhibits a fan-like shape, with the solder splattering into the glass substrate as particles. This phenomenon primarily occurs because the energy density at this level greatly exceeds the ablation threshold of the solder, resulting in a phase explosion or vaporization that causes the solder to splash into the molten glass substrate. Concurrently, the molten glass is pushed downwards, filling the void left by the ablated solder. This process markedly enhances the intermixing of glass and solder, resulting in the formation of a robust weld joint and thus achieving the highest welding strength. As the pulse energy consistently rises to 6μJ, the molten pool's volume expands, and solder particles are splashed to greater heights. Nonetheless, inevitably, pronounced welding defects such as cracks and pores emerge at the interface, resulting in a direct reduction in welding strength. Upon reaching a pulse energy of 8μJ, the SEM image reveals significant cracks at the interface between the glass and the solder, nearly traversing the entire weld zone. Concurrently, the glass exhibits numerous large and small cavities, signifying extensive ablation within its structure. The OM image further illustrates that the femtosecond laser has induced profound modifications to the glass's interior, leading to the creation of high plasma channels within the glass.



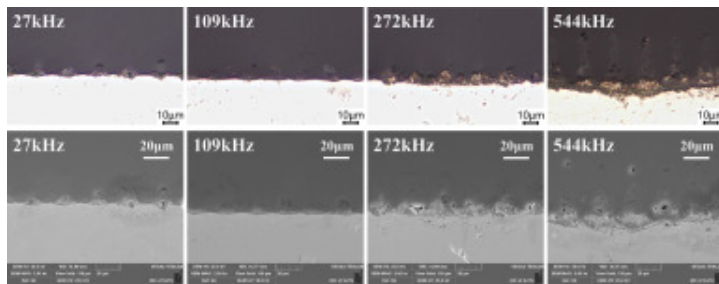
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Fig. 4. Microstructural images of welds captured by OM (upper side) and SEM (lower side) under varying pulse energy.

Within the pulse repetition frequency variable experimental group, the pulse energy is set at $4\mu\text{J}$, with only one single scan being performed. As illustrated in Fig. 3(b), an increase in pulse repetition frequency from 15 kHz to 27 kHz leads to an enhancement in shearing strength. However, a further rise to 109 kHz induces a decline in shearing strength. Within the range of 109 kHz to 1087 kHz, the shearing strength gradually ascends, peaking at 1087 kHz. As illustrated in Fig. 5, at a frequency of 27 kHz, the weld seam exhibits a well-defined hill-like structure, featuring optimal mixing between the solder and glass. This results in robust mechanical anchoring within the welding zone, conferring high welding strength. Conversely, at 109 kHz, the mixing between glass and solder deteriorates, leading to a relatively flat interface that significantly diminishes the mechanical anchoring effect. Additional pulses appear to impact the interior of the glass, creating numerous micro-pores. As the pulse repetition frequency rises to 272 kHz, the molten pool expands considerably, and the interfacial material mixing becomes more homogenous. Nevertheless, the prevalence of defects such as micro-pores becomes more evident. When the frequency reaches 544 kHz, substantial modifications within the glass are visible in OM images, manifesting as pronounced plasma channels. SEM images also reveal excellent material mixing at the interface, and the inherent pore defects at the

interface are relocated to the interior of the glass by the molten pool's flow, thereby endowing the interface with high welding strength.



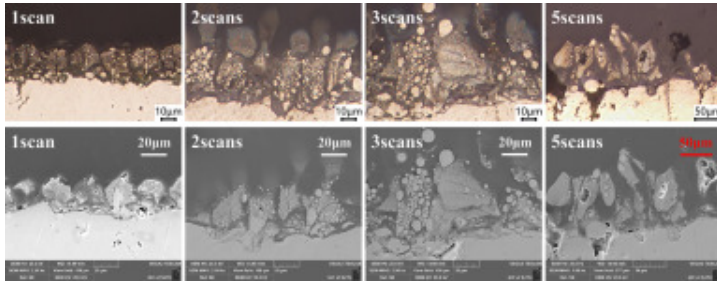
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Fig. 5. Microstructural images of welds captured by OM (upper side) and SEM (lower side) under varying pulse repetition frequency.

Within the number of repeat scans variable experimental group, the pulse energy is set at $4\mu\text{J}$ and the pulse repetition frequency is set at 1087kHz. As depicted in Fig. 3(c), the shearing strength initially experiences a slight increase followed by a marginal decrease as the number of repeated scans rises. In general, changing the number of repeated scans exerts a relatively minor impact on the welding strength. The peak shearing strength of 44.8 Mpa is attained when the number of repeated scans is set to 3. As depicted in Fig. 6, upon a single repetition of the scanning process, the weld's morphology predominantly consists of a material mixing layer, which is created by the spattering of solder particles into the glass. This layer, characterized by a height of approximately $30\mu\text{m}$, can be described as a particle mixing layer. With a second scan, an additional element diffusion layer, exhibiting a lower concentration of solder elements, is superimposed upon the particle mixing layer established by the initial scan. The morphology of this layer resembles a cloud, with its concentration tapering off from bottom to top. Concurrently, it is observed that the particle mixing layer's height has undergone a significant increase, reaching about $50\mu\text{m}$. Upon a third scan repetition, the particle mixing layer's height surges to approximately $80\mu\text{m}$, while the element diffusion layer persists, though its

height remains relatively unchanged. Upon the fifth scan repetition, the particle mixing layer's height attains about $120\mu\text{m}$, yet the element diffusion layer is no longer distinctly visible. Notably, the material interface and the interior of the glass exhibit apparent defects, such as pores and micro-cracks, leading to a reduction in welding strength. In summary, increasing the number of repeat scans effectively augments the height of the material mixing layer, facilitating greater diffusion of the solder's elements into the glass interior.



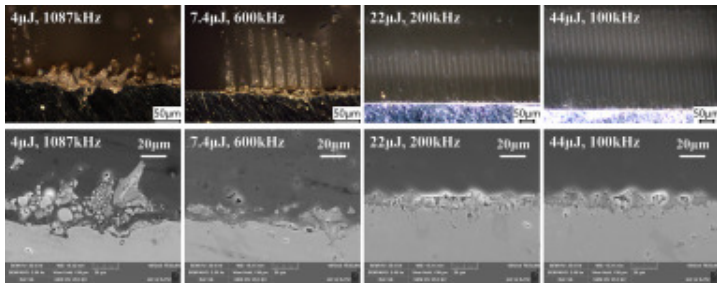
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Fig. 6. Microstructural images of welds captured by OM (upper side) and SEM (lower side) under varying number of repeat scans.

However, excessive scan repetitions can escalate the incidence of welding defects. The underlying mechanism is as follows: with each scan, a portion of the solder droplets or plasma splashes into the molten glass. When subsequent laser scans overlap with the already formed morphology, the metal particle on the upper part of the morphology continues to absorb laser energy, causing further upward splashing. Meanwhile, the portion of the laser that transmits downward acts on the metal at the bottom, promoting further mixing. However, when the metal elements splash to a sufficient height and the laser is severely defocused, the mixing layer no longer continues to grow. At this point, if additional scans are applied, it will cause significant ablation damage to the interface.

In the constant power experimental group, a baseline of 4.4W at 4 μ J, 1087kHz is established, with the power hold constant while the pulse energy and pulse repetition frequency are adjusted. As illustrated in Fig. 3(d), an initial decrease in shearing strength to 23.6 Mpa is observed with increasing pulse energy and decreasing frequency, followed by an increase to 34.9 Mpa. Nevertheless, further increasing the pulse energy result in a trend of diminishing strength. Upon examining the OM images in Fig. 7, it becomes clear that as pulse energy rises and pulse repetition frequency falls, more distinct plasma channels are generated within the glass, and these channels' heights also increases. This phenomenon occurs because the energy density of the femtosecond laser at this point is adequate to excite strong plasma. Shielded by the plasma, the femtosecond laser's position of action progressively ascends, thus forming a moving channel of plasma clusters. Observations from the SEM microscopic morphology of the interface reveal that material mixing at the interface is superior under the parameters of 4 μ J, 1087kHz, whereas it deteriorates under the parameters of 7.4 μ J, 600kHz. A greater number of femtosecond laser pulses are impacting the interior of the glass, yet their contribution to the mixing of the molten pool at the interface is diminished. The weld morphologies at the interface under the parameters of 22 μ J, 200kHz and 44 μ J, 100kHz are quite analogous, both exhibiting a closely packed, small hill-like structure, accompanied by micropores within. Since more femtosecond laser pulses are expended within the glass for the development of plasma channels, only the initial part of the pulses at each weld is utilized for the connection at the interface, leading to inconsistent material mixing at the welds and reduced weld sizes. In summary, increasing the pulse energy leads to a significant rise in the laser intensity along the incident direction. As a result, plasma can be excited at positions far from the interface, markedly enhancing the growth height of the plasma channel. However, more energy is absorbed within the glass, resulting in insufficient melting at the solder interface and a reduced connection strength.



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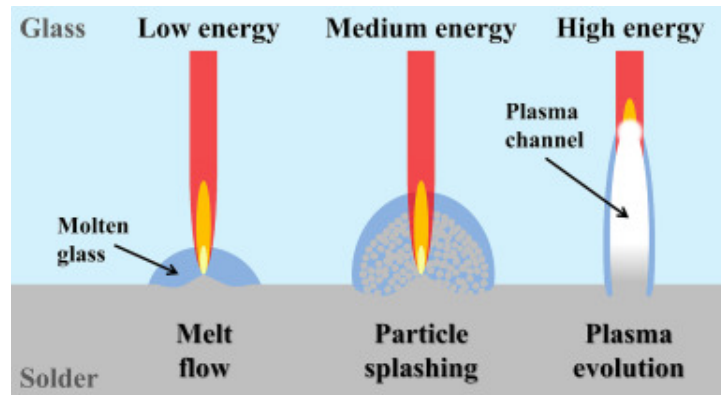
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Fig. 7. Microstructural images of welds captured by OM (upper side) and SEM (lower side) under the condition of maintaining a constant power of 4.4W, while simultaneously varying the pulse energy and pulse repetition frequency.

3.2. Material interface element migration mechanism and welding strategy optimization

Based on the experimental results presented, it is evident that the mixing mechanisms of interface materials at varying energy levels, as depicted in Fig. 8, exhibit distinct characteristics. At low energy inputs, the mixing of glass and solder is predominantly governed by melt flow, which is characterized by the emergence of convex-concave structures at the interface. This way of material mixing can form a good mechanical anchorage connection, with fewer welding defects. With a moderate energy input, phenomena such as phase explosion and vaporization of the metal solder become prevalent, leading to the mixing of glass and solder through particle splashing. Specifically, this involves solder particles splashing into the molten glass matrix, while the molten glass fills the cavities left by the ablated solder. This way of material mixing has a high welding strength, with a relatively uniform mixing of materials and fewer welding defects. However, when the energy input is high, it triggers intense plasma generation in both metal and even glass. As the plasma absorbs laser energy and expands upwards, a plasma channel is established. At this point, glass and solder remain

connected through melt mixing at the interface, while within the plasma channel, more microscopic elemental mixing occurs. This way of material mixing can lead to increased defects such as pores or micro-cracks at the material interface and within the glass, leading to a decrease in welding strength.



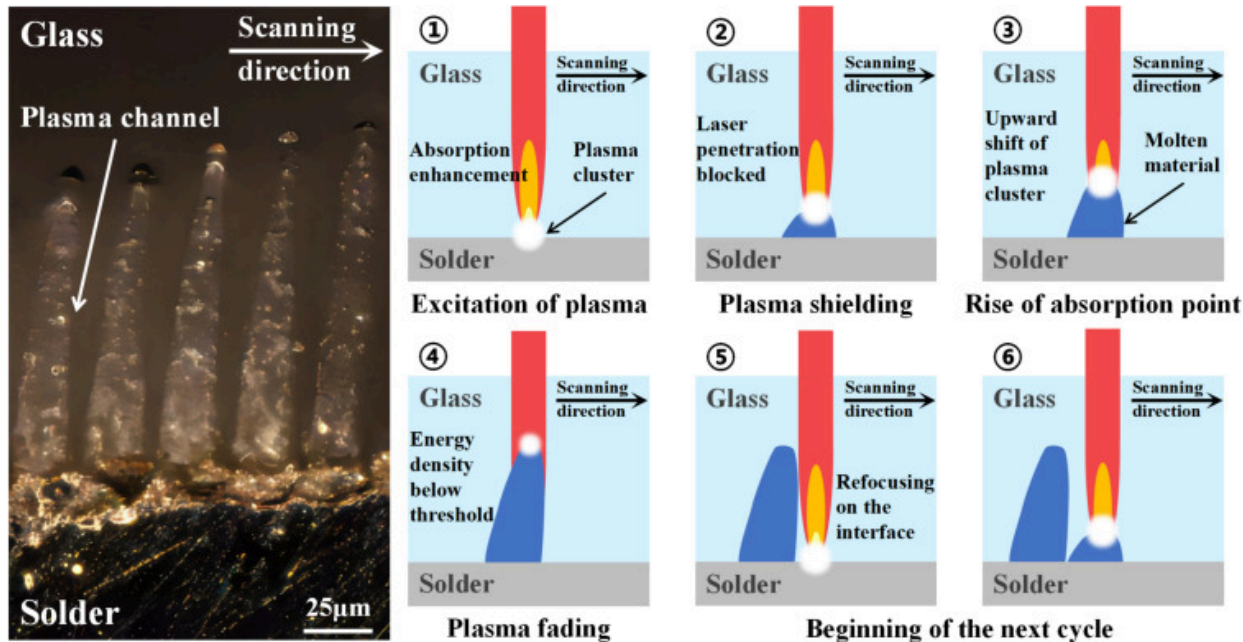
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Fig. 8. Interface material mixing mechanisms under various energy conditions.

The formation of a plasma channel, while potentially weakening the weld's strength, breaks through the conventional method of material mixing in welding. It offers a novel strategy for the migration of base materials into the interior of glass materials. Specifically, for the pressing issue of thermal fatigue in glass-metal welding, it facilitates the creation of macroscopic elemental transition layer, thereby enabling the matching of interface properties between two materials with significant differences. To effectively utilize the plasma channel for the creation of elemental transition layer, an analysis of the plasma evolution mechanism during the welding process is conducted, as depicted in Fig. 9. Initially, under the action of high energy femtosecond laser pulses, plasma is excited at the interface. The plasma clusters exhibit a high absorption for femtosecond laser pulses, resulting in a robust absorption effect on subsequent laser pulses, which in turn shields the downward propagation of subsequent pulses [28]. Since the upper region of the

plasma cluster absorbs laser energy preferentially, new plasma is continually excited on its upper surface, prompting the cluster to ascend. As the plasma cluster ascends to a sufficiently elevated position, the laser energy density at that point diminishes to a level insufficient to excite additional plasma, leading to a marked reduction in plasma intensity and signaling the plasma channel has reached its peak [29]. Since the laser is scanned for processing, as the laser spot transitions to a new location, fresh plasma is excited at the interface, initiating the growth of the subsequent plasma channel.



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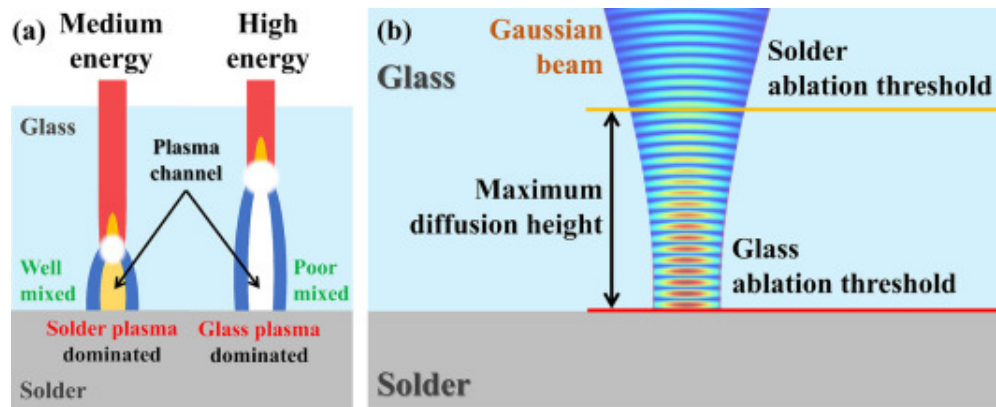
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Fig. 9. The evolution mechanism of plasma throughout the welding process.

Generating a plasma channel is crucial, yet more crucially, this plasma channel must serve as a pathway for the diffusion of solder elements into the interior of the glass.

Consequently, the challenge lies in transporting these solder elements to a sufficiently elevated position. As illustrated in Fig. 7, when the pulse energy reaches a sufficiently

high level, remarkably tall plasma channels can develop within the glass. However, these channels lack substantial solder element composition, thereby failing to accomplish the mixing of solder and glass elements. Upon analysis, it is deduced that the type of plasma is intimately tied to the magnitude of the pulse energy, as depicted in Fig. 10(a). The absorption of laser energy by glass is directly related to the ionization process, making the ablation threshold of glass nearly equivalent to the plasma formation threshold. In contrast, for most metals, the plasma formation threshold is typically slightly higher than the ablation threshold [30]. Due to a lack of sufficient literature on the study of metal plasma formation thresholds, which results in incomplete data, the ablation threshold is used for qualitative analysis to clarify the mechanism. When the pulse energy is moderate (solder ablation threshold < femtosecond laser energy density < glass ablation threshold), the plasma generated is predominantly composed of solder elements. At this time, the solder plasma ascends within the molten glass, facilitating thorough mixing of solder and glass elements until the excitation of the solder plasma ceases. Conversely, when the pulse energy is excessively high (femtosecond laser energy density > glass ablation threshold), both solder and glass plasmas can be excited at the interface. Nevertheless, since laser energy absorption primarily occurs above the plasma cluster, the subsequently generated plasma is predominantly comprised of glass elements. Under the combined influence of the plasma's strong absorption and incubation effects, the ablation threshold of the glass progressively decreases [31]. Consequently, even at highly elevated positions (significant defocus amounts), glass plasma can still be excited. Therefore, to obtain a plasma channel dominated by solder elements, it is imperative to ensure that the femtosecond laser energy density remains below the ablation threshold of the glass. This step is crucial to prevent the excitation of intense glass plasma at the interface.



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Fig. 10. (a) Schematic diagram of plasma types produced at different laser energies; (b) Schematic diagram of the influence mechanism of the diffusion height of solder elements.

Regarding how to maximize the diffusion height of solder elements into the glass, a mechanism that affects the height has been proposed, as depicted in Fig. 10(b). For a Gaussian beam, the energy density peaks at the beam waist and diminishes as it ascends from this point. The ablation threshold of the solder marks the apex of plasma channel growth; thus, the greater the laser energy density, the taller the plasma channel can extend. Simultaneously, to prevent the excessive excitation of plasma within the glass, the energy density at the beam waist must remain below the glass's ablation threshold. Ultimately, the theoretical maximum height of solder element diffusion is dictated by the beam defocus amount that corresponding to the difference between the solder ablation threshold and the glass ablation threshold.

In order to maximize the gap between the ablation threshold of glass and solder, thereby enhancing the maximum diffusion height, a comparison is made of the melting points and ablation thresholds of various materials, as illustrated in Table 1. The table reveals that tin (Sn) possesses the lowest melting point and ablation threshold, significantly lower than those of other metal elements. Consequently, this research has chosen tin-

based solder as the material to be inserted between glass and metal substrates. The employment of tin-based solder offers several benefits: (1) It exhibits excellent bonding characteristics with a wide range of metals, broadening the spectrum of metals that can be joined; (2) With its very low melting point, tin remains molten during high-power femtosecond laser welding, which compensates for the expansion and distortion of glass, markedly diminishing welding thermal stress and preventing glass cracking. (Conventional ultrafast laser glass-metal welding power is typically about 1–2W [32]); (3) The ablation threshold of tin is notably low, creating a substantial gap with the ablation threshold of glass, thus permitting a greater element diffusion height.

Table 1. Melting points and ablation thresholds of various materials.

Material	Melting point (°C)	Ablation threshold (J/cm ²)
SiO ₂	≈1710	6.23 (200fs, 1030nm) [33]
Sn	232	0.042 (65fs, 800nm) [30]
Fe (304 Stainless steel)	1454	0.39 (190fs, 1026nm) [34]
Al (6061 Aluminum alloy)	582	0.43 (190fs, 1026nm) [34]
Cu	1083	1.06 (130fs, 800nm) [35]
Ag	961	1.51 (100fs, 800nm) [36]
Ti	1668	0.074 (40fs, 800nm) [37]

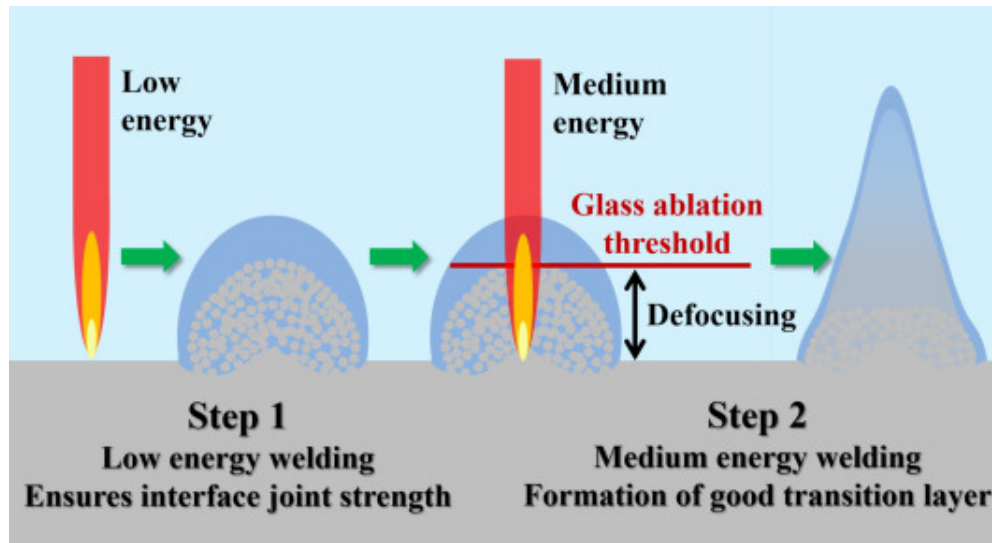
To improve the thermal fatigue resistance of glass-metal welded samples while preserving robust welding strength, an optimized femtosecond welding strategy is introduced, as depicted in Fig. 11. First, the optimal welding strength parameters obtained from parameter experiments (4μJ, 1087kHz) are used to connect the glass and solder interface to ensure high welding strength. The second step is to build upon this foundation by applying a higher energy input (6μJ, 1087kHz) during the welding process.

This step aims to create a plasma channel and establish a higher material transition layer, resulting in a minimal gradient change in the interface element concentration, thereby diminishing stress during thermal cycling. The choice of 6μJ energy is established based on the ablation threshold of quartz glass, as calculated by Eqs. (1), (2), (3). In these equations, $\omega(l)$ represents the beam radius at a distance l away from the focal plane, ω_0 is the beam radius at the focal plane, Z_R is the Rayleigh length of the laser, λ is the laser wavelength, E is the pulse energy, and $F(l)$ is the laser energy density. Utilize the weld profile height of 30μm acquired from the first scan as the defocus amount for the second scan. This implies that the laser energy density at a height of 30μm above the interface matches the ablation threshold of quartz glass. Consequently, the pulse energy to be implemented is 5.96μJ, thus, this research adopts 6μJ as the welding parameter for the second step.

$$\omega(l) = \omega_0 \sqrt{1 + \left(\frac{l}{Z_R}\right)^2} \quad (1)$$

$$Z_R = \frac{\pi \omega_0^2}{\lambda} \quad (2)$$

$$F(l) = \frac{2E}{\pi \omega(l)^2} \quad (3)$$



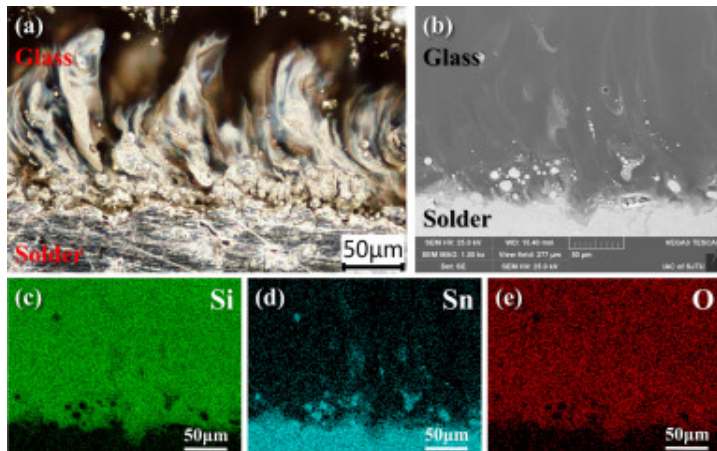
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Fig. 11. Optimized welding strategy for constructing the transition layer of interface materials.

3.3. Characterization of the physical and chemical properties of welds and thermal fatigue resistance testing

Upon employing the optimized welding strategy, the cross-sectional microstructure of the weld is examined, as depicted in Fig. 12. Fig. 12(a,b) are OM and SEM images of the weld seam respectively. It is evident that at the interface between the glass and the solder, a favorable mixing and connection are achieved through the way of melt flow and particle splashing. Above the material interface, solder elements are conveyed into the interior of the glass matrix via plasma channels, resulting in a gradient transition of solder and glass elements from the bottom upwards. Fig. 12(c-e) illustrate the EDS mapping images of the weld, revealing a significant migration of the primary solder component Sn, into the glass matrix.



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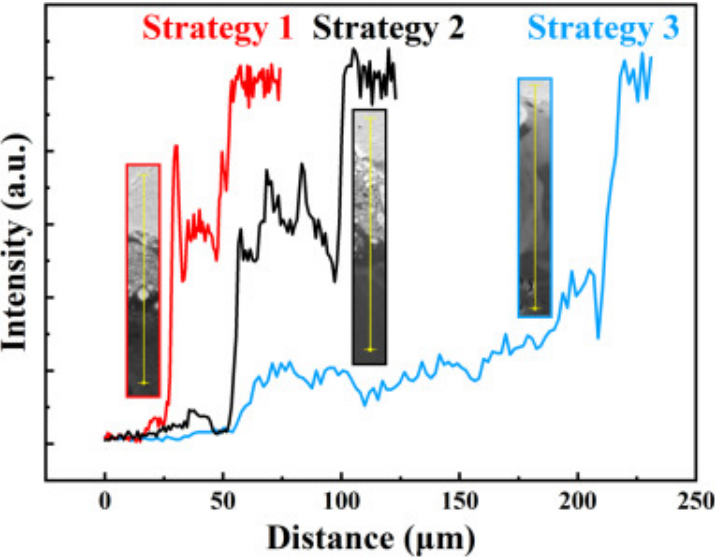
Fig. 12. Optimized weld microstructure characterization. (a) OM image; (b) SEM image; (c) EDS mapping image of Si element; (d) EDS mapping image of Sn element; (e) EDS mapping image of O element.

To more vividly demonstrate the efficacy of the optimized welding strategy in creating a superior material gradient transition layer, EDS line scanning is employed to contrast the Sn element diffusion curves across three distinct welding strategies, and the parameters of the three strategies are shown in [Table 2](#): Strategy 1 entails a single scan at 4µJ and 1087kHz; Strategy 2 involves three scans at 4µJ and 1087kHz; while Strategy 3 combines a single scan at 4µJ and 1087kHz with another scan at 6µJ and 1087kHz. The comparative outcomes are depicted in [Fig. 13](#). Observing the figure, Strategy 1 yields a diffusion distance of approximately 70µm, with the Sn element concentration exhibiting a stepped escalation. Strategy 2 extends the diffusion distance to roughly 120µm, maintaining a stepped increase in Sn element concentration. Strategy 3 achieves a diffusion distance of approximately 230µm, and although the change in Sn element concentration at the solder-glass interface remains notably abrupt, the concentration within the glass reveals a remarkably smooth gradient variation. To emphasize the significant breakthrough of this

study in constructing the transition layer at the glass-metal interface, the sizes of the transition layers in other ultrafast laser glass-metal welding literature are compared, as shown in Table 3. As can be seen, the size of the transition layer is dozens of times larger than the average found in the literature. In summary, the optimized welding strategy markedly improves both the extent and the concentration gradient of element diffusion.

Table 2. The laser parameters of three welding strategies.

Strategy	Single pulse energy	Pulse Repetition Frequency	Number of scans
Strategy 1	4μJ	1087kHz	1
Strategy 2	4μJ	1087kHz	3
Strategy 3	4μJ	1087kHz	1
	6μJ	1087kHz	1



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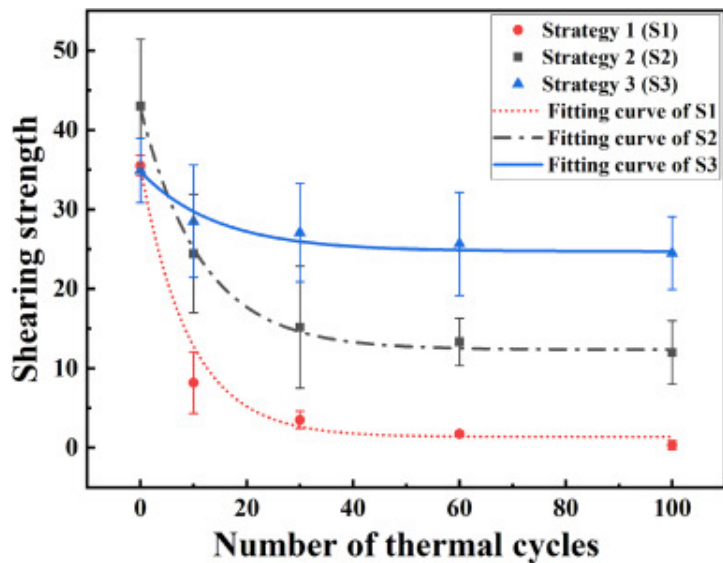
Fig. 13. Diffusion curves of Sn element under three welding strategies.

Table 3. Comparison of elemental transition layer sizes at the interface of ultrafast laser glass-metal welding.

References	Material	Elemental transition layer size
This study	Quartz glass - Tin	230 μ m
Reference1 (2008) [38]	Non-alkali glass - Copper	2 μ m
Reference2 (2016) [18]	Quartz glass - Aluminum	10 μ m
Reference3 (2024) [39]	Sapphire - Invar36 alloy	2 μ m

Thermal fatigue resistance testing on welded specimens is performed within a rapid temperature change test chamber. The specific test parameters are as follows: temperature range of -40°C to 150°C ; stabilization time of no less than 0.5h; and a temperature change rate of $15^{\circ}\text{C}/\text{min}$. The variation curve of shearing strength with the number of thermal cycles under three different welding strategies is illustrated in Fig. 14. Based on the test results, it is evident that due to the small element diffusion distance and large element concentration gradient associated with Strategy 1, during the process of high and low temperature cycling, it experiences greater thermal stress. Consequently, the attenuation of shearing strength under thermal cycling occurs very rapidly. After just 30 thermal cycles, the shearing strength has decreased by 90.1%, and after 100 thermal cycles, the attenuation of shearing strength exceeds 95.1%. In contrast, Strategy 2 exhibits an improved element diffusion distance and concentration gradient, which results in lower thermal stress during the process of thermal cycling, thereby enhancing resistance to thermal fatigue. After 30 thermal cycles, the shearing strength has decreased by 64.7%, and after 100 thermal cycles, the attenuation of shearing strength exceeds 72.1%. Among the three strategies, due to the largest element diffusion distance and the smallest concentration gradient, Strategy 3 exhibits the greatest resistance to thermal fatigue.

After 30 thermal cycles, the shearing strength has decreased by only 22.4%, and after 100 thermal cycles, the attenuation of shearing strength remains at only 29.8%.

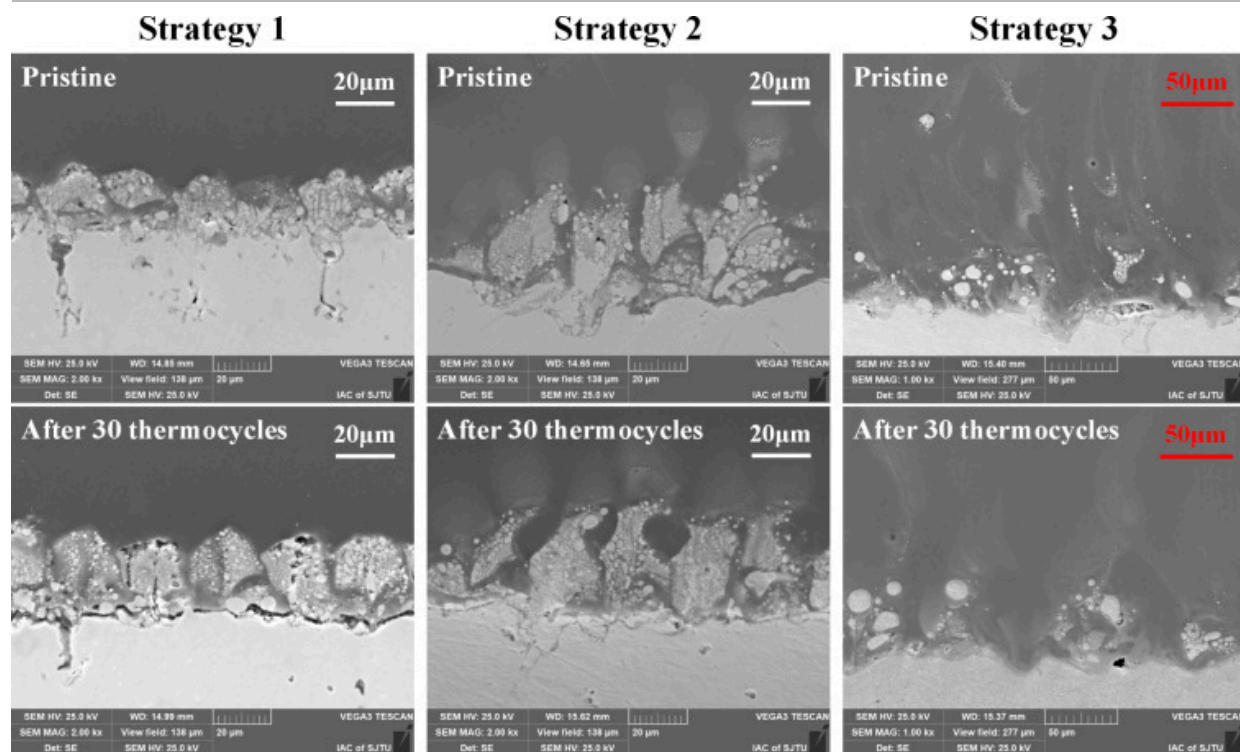


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Fig. 14. Shearing strength variation curve with number of thermal cycles under three welding strategies.

The transformation of weld morphology after 30 thermal cycles under three different welding strategies is depicted in Fig. 15. Strategy 1 resulted in a pronounced crack at the glass-solder interface after 30 thermal cycles, which nearly spanned the entire welding area, leading to a significant reduction in welding strength. Strategy 2 also displayed micro-cracks at the glass-solder interface, but these cracks are narrower and have not fully propagated, preserving partial connection between the glass and solder. Strategy 3 does not exhibit any apparent cracks or defects in the weld after 30 thermal cycles, thus preserving a high level of welding strength.

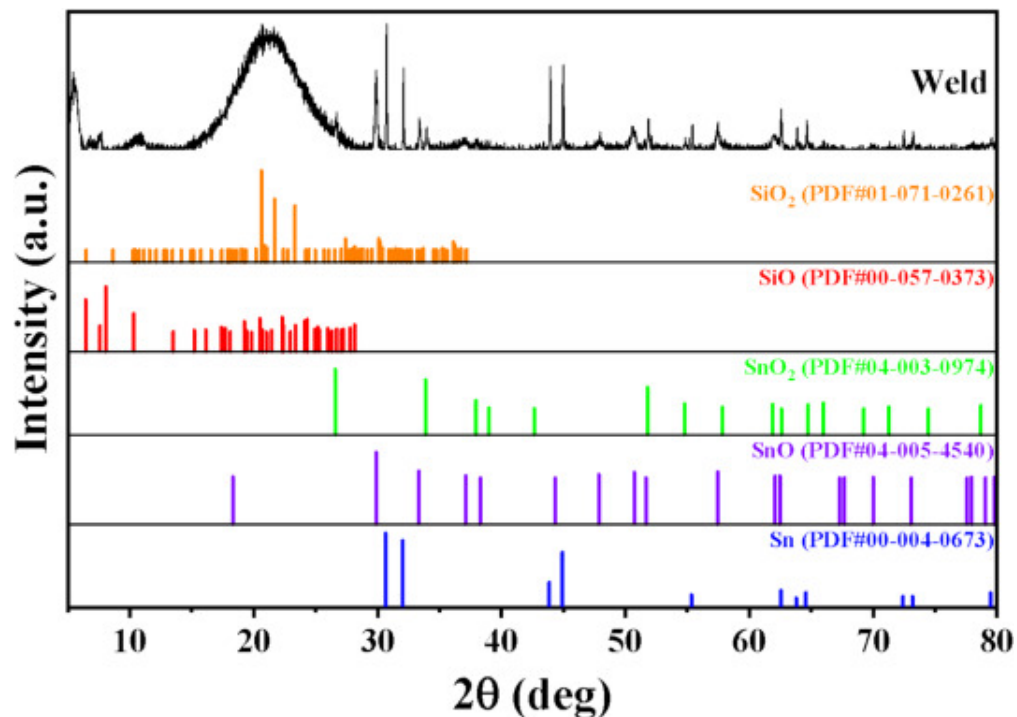


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Fig. 15. The transformation of weld morphology after 30 thermal cycles under three different welding strategies.

Then the material composition of the welding sample by using strategy 3 is analyzed by XRD. Experimental XRD patterns of fracture surface at metal substrate of the weld sample are shown in Fig. 16. The uppermost black curve represents the XRD pattern of the weld area obtained during the experiment, whereas the colored lines correspond to data from the PDF card database of XRD to identify the source of diffraction peaks. Analysis of the test outcomes reveals that the material phases potentially existing at the interface post-welding encompass SiO_2 , SiO , SnO_2 , SnO and Sn , findings that are largely in agreement with existing literature reports [40].



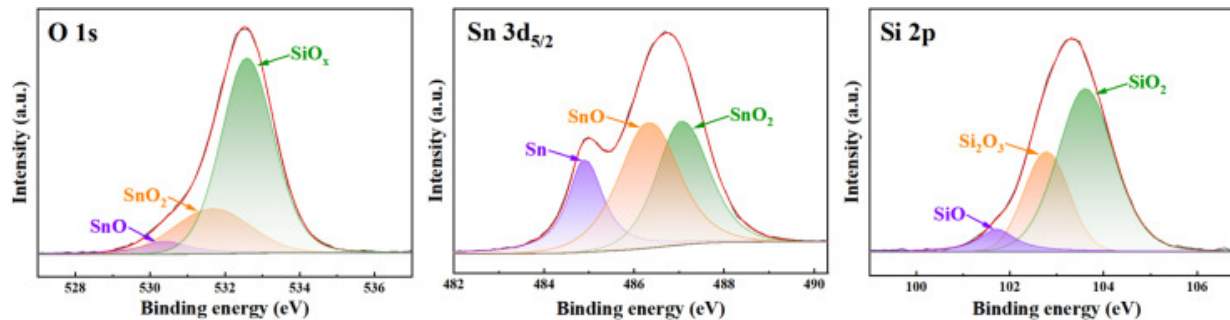
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Fig. 16. Experimental XRD patterns of fracture surface at metal substrate of the weld sample.

The material composition within welds is further elucidated using high-resolution XPS, as depicted in Fig. 17. All spectral peaks are calibrated to the standard carbon C 1s binding energy of 284.8eV to mitigate any extraneous disturbances (such as surface states, charge effects, etc.). A shift in the binding energy of metals will be observed in a single compound phase, such as metal oxides, if they are to become integrated into an inorganic complex with distinct phases [41]. The O 1s spectra of the weld is deconvoluted into three peaks (530.41 eV, 531.68eV and 532.60eV), attributed to the oxygen in metal oxides (SnO and SnO₂) and silicon oxides respectively [42,43]. The Sn 3d_{5/2} spectra of the weld exhibit three peaks (484.89eV, 486.30eV and 487.05eV), corresponding to different Sn oxidation

degrees of 0 (metal), +2 (SnO) and +4 (SnO₂) [40]. The Si 2p spectra of the weld have been resolved into three peaks (101.72 eV, 102.79 eV and 103.62 eV), which can be identified as Si²⁺ (SiO), Si³⁺ (Si₂O₃) and Si⁴⁺ (SiO₂) respectively [44,45].



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Fig. 17. High-resolution XPS spectra of O 1s, Sn 3d_{5/2}, and Si 2p on the fracture surface at metal substrate of the weld sample.

Furthermore, the application of OM and SEM for microscopic observation of the weld morphology, coupled with the use of XRD and XPS to analyze the phases at the weld seam, has essentially elucidated that the bond between glass and metal under the optimized welding strategy is established via the synergistic effect of three distinct methods: (1) mechanical anchoring achieved through the flow of molten material; (2) chemical bonding resulting from chemical reactions that lead to the creation of new phases; and (3) intermolecular forces arising from the mixing of materials and the diffusion of elements. Consequently, the optimized welding strategy achieves a high level of welding strength.

4. Conclusion

This study concentrates on the thermal fatigue resistance requirements of glass-metal femtosecond laser weld seam. It conducted experimental research on the effects of

parameters on the welding strength and microstructure of glass-metal joints, elucidating the elemental migration mechanism at the interface of heterogeneous materials. The primary findings of this study are summarized as follows:

- (1) A novel welding strategy, which involves the creation of plasma channels using a femtosecond laser, is proposed to form an elemental concentration gradient transition layer with a thickness exceeding 200 μ m at the interface of heterogeneous materials.
- (2) Thanks to the matching of thermal properties at the material interface, the thermal fatigue resistance of glass-metal weld is significantly improved. After 100 thermal cycles ranging from -40°C to 150°C , in comparison to samples without a gradient transition layer, the attenuation of welding strength significantly decreased from 95.1% to 29.8%.
- (3) A significant amount of elemental intermixing occurs at the interface, resulting in the creation of several new phases. Additionally, it is clarified that the bond between glass and metal is formed through a combination of mechanical anchoring, chemical bonding, and intermolecular forces, which contributes to its high welding strength.

CRediT authorship contribution statement

Haodong Ren: Writing – original draft, Validation, Software, Investigation, Formal analysis, Data curation. **Hong Shen:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work is supported by Natural Science Foundation of Shanghai (No.[24ZR1436000](#)).

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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